

Influence of dietary protein supplements on the formation of bacterial metabolites in the colon.

BACKGROUND: To evaluate the influence of increased dietary protein intake on bacterial colonic metabolism in healthy volunteers. **METHODS:** Short chain fatty acids, ammonia, and volatile organic compounds in faecal samples, and phenols in the urine of five volunteers were measured after one week of basal nutrient intake and after one week of a diet supplemented with a protein rich food (Fortimel; Nutricia, Zoetermeer, The Netherlands). Paired t tests and factor analysis were used for statistical analysis. **RESULTS:** Total energy and resistant carbohydrate intake remained unchanged in each study period. The percentage energy intake delivered as dietary protein, increased significantly (from 15.4% to 23.8%; $p = 0.007$) during supplement intake. A significant increase in faecal ammonia ($p = 0.002$), faecal valeric acid ($p = 0.02$), and urinary p-cresol ($p = 0.04$) was noted during supplementary protein intake. A total of 120 different volatile compounds were isolated from the faecal samples of which 10 increased significantly during dietary protein supplementation. The change in volatile pattern, especially for S containing metabolites, was clearly shown by a factor analysis model which made a distinction between the two dietary regimens for all volunteers. **CONCLUSION:** An increase in dietary protein leads to altered products formation by colonic metabolism, mainly reflected by an increase in faecal ammonia, faecal volatile S substances, and urinary p-cresol.